

CLASH OF THE CLAWS

RIT Team Gladiolus On-device Personal Recommendation System

Members:

Omkar Kulkarni

Harsha Prasad

Pragnashri V

Palak Ojha



PROBLEM STATEMENT

Problem Statement : Current wearables rely on cloud processing to deliver health insights, creating privacy risks, latency, and one-size-fits-all recommendations that fail individual users. There is a clear need for an on-device solution that processes biometric data locally, ensuring real-time, private, and personalized health guidance without cloud dependency.

IDEA

We build a fully on-device GenAI health coach for Samsung wearables with zero cloud dependency and complete data privacy. It fuses biosignals like heart rate, HRV, sleep, and activity into real-time insights, uses local RAG memory for personalization, and runs a quantized LLM for sub-200ms coaching—delivering continuous, contextual health guidance instead of delayed or isolated metrics



WHY THIS THEME

Modern wearables collect rich biometric data, yet most intelligence still relies on cloud AI. With advances in Edge AI and Samsung NPUs, on-device GenAI is now feasible—enabling private, real-time personalized wellness coaching directly on wearables.



TARGET USERS/PERSONAS

Students — Stress and sleep management

Professionals — Fatigue and lifestyle monitoring

Fitness Users — Recovery and performance optimization

Elderly Users — Private continuous health monitoring



WHY IT MATTERS

Current wellness apps offer generic insights with limited personalization.

Cloud-based systems introduce latency, privacy risks, and internet dependency.

Real-time personalized coaching can improve preventive healthcare and daily wellness habits.

CURRENT SOLUTIONS AND GAPS

EXISTING SOLUTIONS

SAMSUNG

SAMSUNG HEALTH/FITBIT

Good sensor tracking, but mostly rule-based insights with limited personalization.



APPLE HEALTH ECOSYSTEM

Strong ecosystem integration, but advanced intelligence heavily depends on cloud processing.

OURA

WHOOOP/ OURA RING

Provides recovery analytics, but lacks fully private on-device GenAI reasoning.



TINY ML-BASED WEARABLES

Efficient on-device sensing, but limited to narrow tasks like step counting or anomaly detection.



CLOUD AI BASED ASSISTANTS

Powerful models, but suffer from latency, privacy concerns, and internet dependency.

LIMITATIONS



ISOLATED DATA STREAMS

Signals are analyzed separately, lacking unified cross-modal understanding and context.



CLOUD DEPENDENCE

Relies heavily on cloud → adds latency, no offline use, and privacy risks.



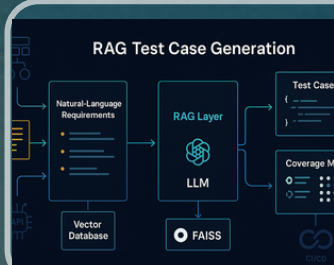
MODEL AND ENERGY CONSTRAINTS

Limited on-device resources; LLMs require heavy optimization to run locally.



PRIVACY/PERSONALIZATION GAP

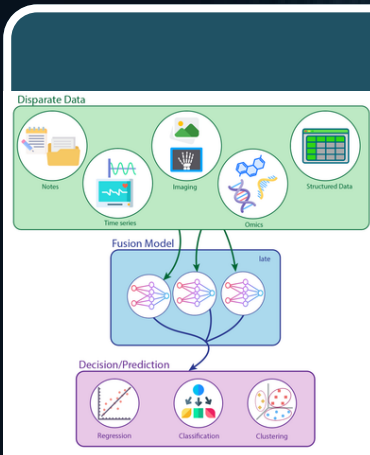
Mostly generic insights; weak personalization and no full on-device privacy.



RAG FEASIBILITY

On-device retrieval memory (RAG) is rare due to storage, compute limits, and limited frameworks.

SOLUTIONS AND NOVELTY



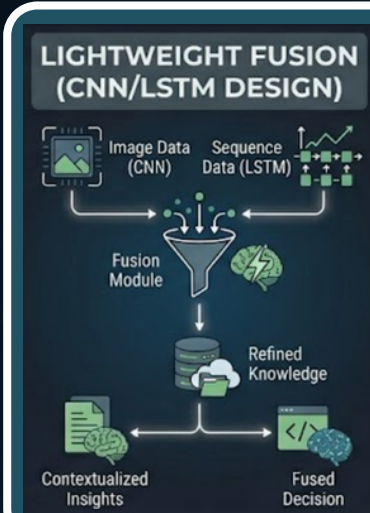
MULTI-MODAL FUSION (LLM BASED)

LLM fuses multi-sensor data via prompts for unified understanding.



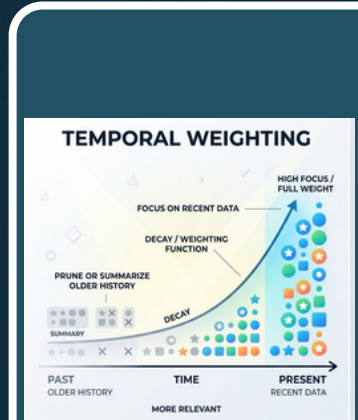
QUANTIZATION STRATEGIES

4–8 bit quantization enables efficient on-device LLM inference.



LIGHTWEIGHT FUSION ALGORITHMS

Small CNN/LSTM extract features, converted into prompts/embeddings.



TEMPORAL WEIGHTING

Focus on recent data; prune or summarize older history.



LOCAL MEMORY (RAG) DESIGN

On-device vector DB stores history; retrieves context for LLM.

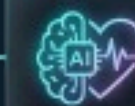


PERSONALIZATION

Learn user baselines; adapt responses using profile and history.



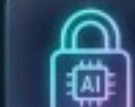
Our Novel AI Health System



Adaptive Coaching
Real-time, context-aware health guidance



Continual Personalization
Learns and adapts over time



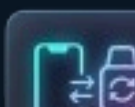
Privacy-First AI
100% on-device, no data leakage



Multimodal Intelligence
Fuses HR, sleep, activity signals



Explainable Insights
Clear reasoning behind every suggestion



Offline Updates
Seamless updates without internet



Energy-Aware AI
Optimized for power and efficiency

Idea

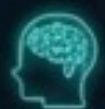
Demo/Product Walkthrough

Screen/Flow

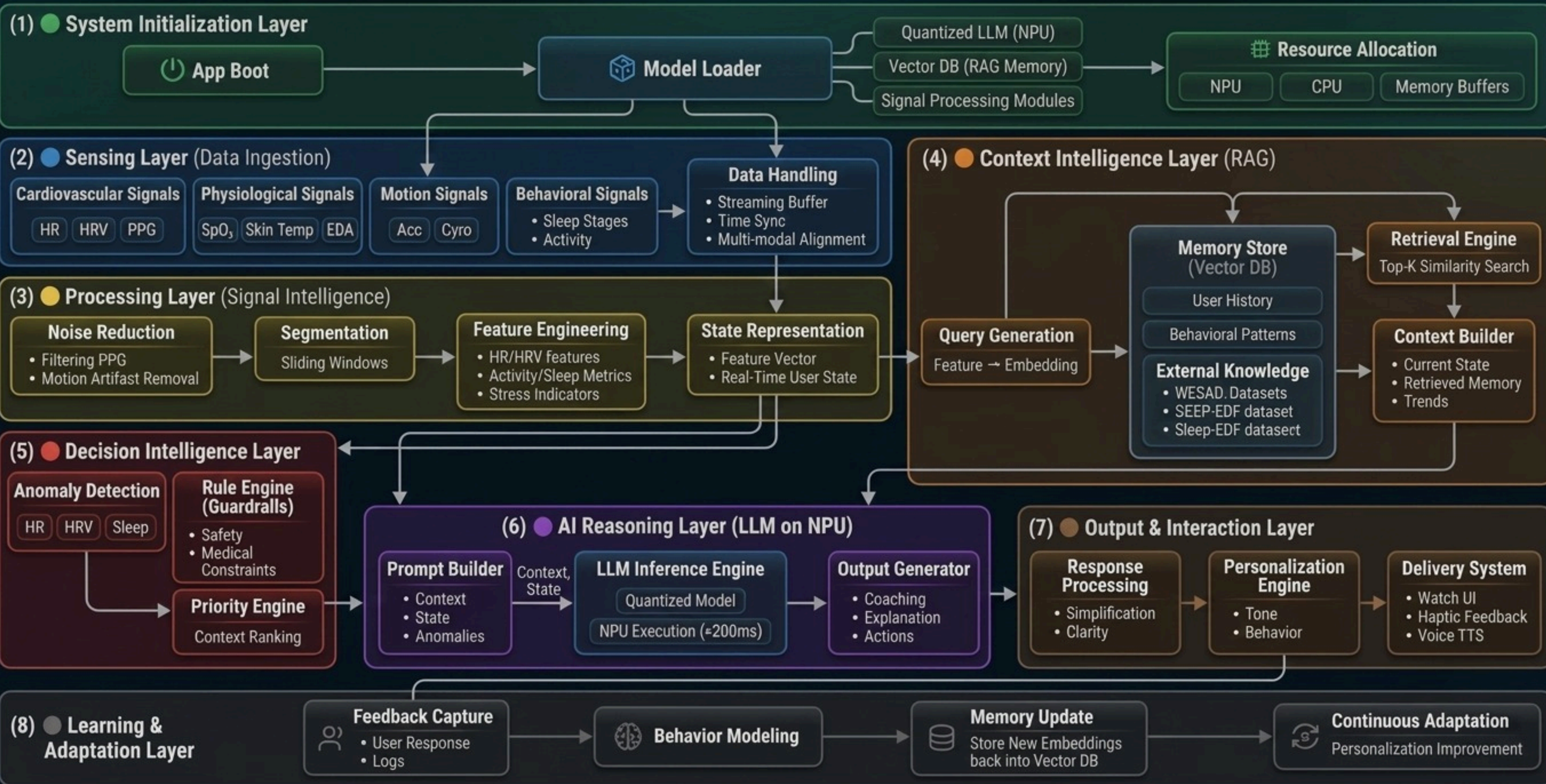
How User Interacts

Key Moments

This is the most important
Slide



ON-DEVICE GENAI HEALTH COACH SYSTEM ARCHITECTURE & DATA FLOW



AI/ML

OpenClaw,
LangGraph,
PyTorch, RAG+FAISS
OpenRouter/Ollama



Frontend

React



Tech Stack

Device/Wearable

Samsung SDK,
NNAPI, TFLite



Backend

FastAPI (Python)



Project Budget

Cost categories

Dev effort	Compute /Cloud	Models & licences	Tooling & SDKs	Hardware / Sensors	Misc & QA
Low: Free Med: ₹2.5L High: ₹8L	Low: Free Med: ₹40k High: ₹1.6L	Low: Free Med: ₹80k High: ₹4L	Low: Free Med: ₹80k High: ₹8L	Low: ₹40k Med: ₹65k High: ₹1L	Low: Free Med: ₹45k High: ₹4L

Low estimate ₹1.2L – 1.6L

- 1 engineer on-device
- Open-source models
- Free cloud tier
- 1 dev device only

Median estimate ₹5L – 8L

- 2–3 engineers
- Standard cloud mode
- Premium dev tools
- 2 test devices

High estimate ₹12L – 25L+

- AI engineers team
- Multi-modal licences
- Enterprise cloud
- 8+ test devices

Total (approx)
₹1.2L – ₹25L+

Scope
decision

MVP / Low

Full / High

Start at
₹1.2L

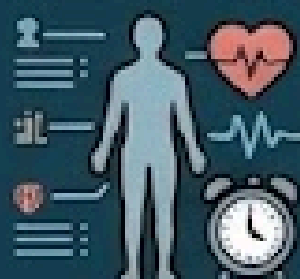
Scale to
₹25L+

IMPACT AND USE CASES

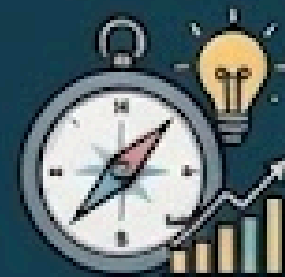
IMPACT



- **Privacy-First Healthcare**
Sensitive biometric data never leaves the device.



- **Preventive Wellness**
Early detection of stress, fatigue, and unhealthy patterns.



- **Personalized Guidance**
Context-aware recommendations tailored to individual baselines.



- **Real-Time Assistance**
Sub-200ms coaching enables instant health feedback.



- **Accessible AI Wellness**
Continuous intelligent coaching directly from wearable devices.

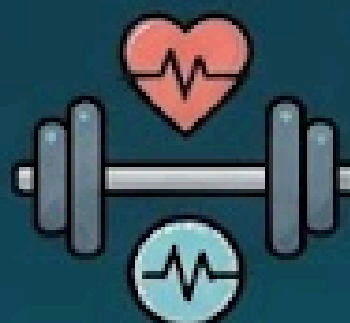
REAL-LIFE USE CASES



- **Student Burnout Detection**
Detects elevated stress + poor sleep during exams and recommends recovery actions.



- **Workplace Wellness**
Identifies prolonged inactivity and rising stress during work hours.



- **Fitness Recovery Optimization**
Analyzes HRV, activity, and sleep quality to adjust workout intensity.



- **Elderly Safety Monitoring**
Detects abnormal heart-rate patterns or falls and triggers immediate alerts.

SCALE POTENTIAL



- **Samsung Ecosystem Expansion**
Can extend across Galaxy Watches, Rings, Phones, and Buds.



Wellness



Fitness



Healthcare



Elder Care

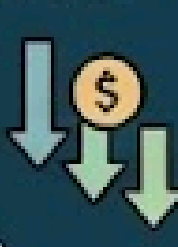
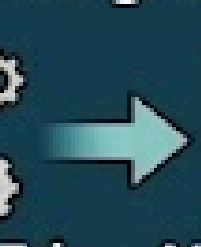
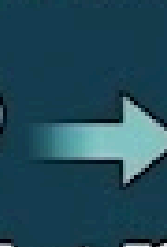
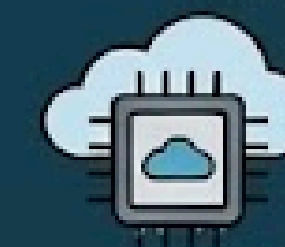
Cross-Industry Applications



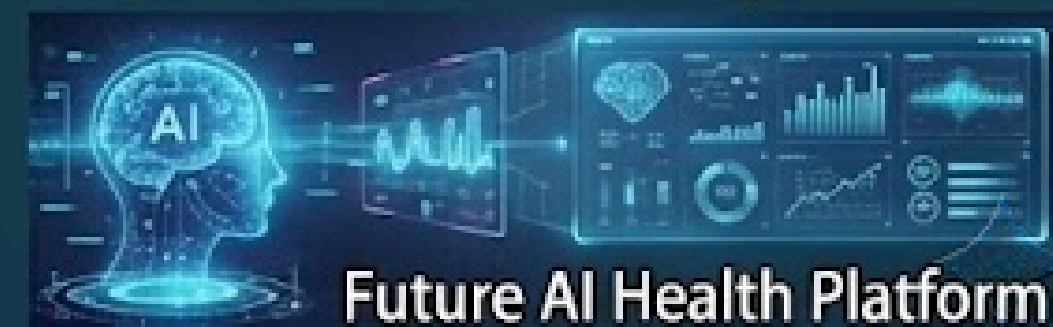
Wearable core



OEM & Enterprise Integration

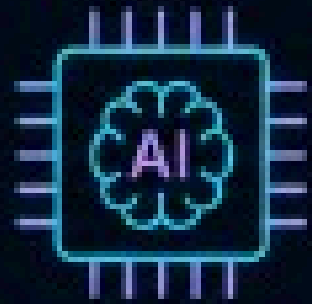


Cost-Efficient Edge AI



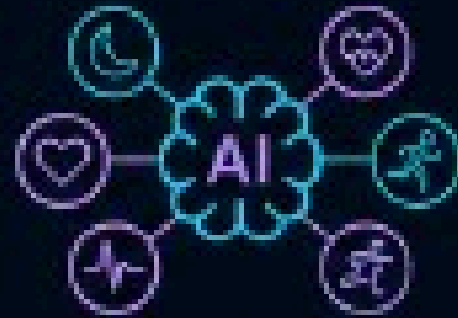
Future AI Health Platform

WHY THIS IDEA STANDS OUT (MOAT)



Fully On-Device GenAI

Runs entirely on-device, zero internet dependency. Complete data privacy.



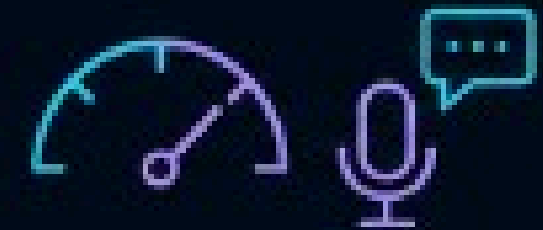
Multimodal Intelligence, Not Isolated Metrics

Fuses HR, HRV, sleep, activity, and behavioral patterns for unified contextual understanding.



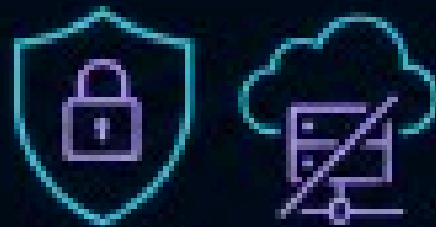
Personalized Long-Term Memory (Local RAG)

Continuously learns from user history through on-device retrieval memory. Coaching that evolves.



Real-Time Contextual Coaching

Adaptive health guidance in under 200ms based on live biometric conditions.



Privacy as a Core Architecture

All sensor data, embeddings, and inference remain local. Eliminates cloud exposure risks.



Edge AI Optimized for Wearables

Combines TinyML, quantized LLMs, and efficient pipelines within wearable hardware limits.



Foundation for Future Samsung AI Ecosystem

Scales across Galaxy devices to create a unified intelligent wellness ecosystem.

DEMO LINK

GitHub repo link <https://github.com/Omkar-S-Kulkarni/samsung>

FEASIBILITY

AND

VIABILITY

Sensor Integration Ready

Real-time multi-modal data via
Samsung Health SDK

On-Device Processing

Lightweight preprocessing +
TinyML within wearable
limits

Quantized LLM Deployment

1B–3B models with 4-bit
quantization on NPU

Local RAG Memory

Compact vector storage
enables personalization

Real-Time Execution

<200ms latency via optimized,
event-driven inference



Strong Market Need

Real-time, personalized,
privacy-first coaching gap

Clear Differentiation

LLM + multimodal fusion + on-
device RAG

Privacy-First Advantage

Fully on-device, no data
leakage

Scalable & Cost-Efficient

No cloud → low cost, easy
scaling

High Product Potential

Samsung Health integration +
OEM licensing